

Network Connectivity Testing

Network Topology

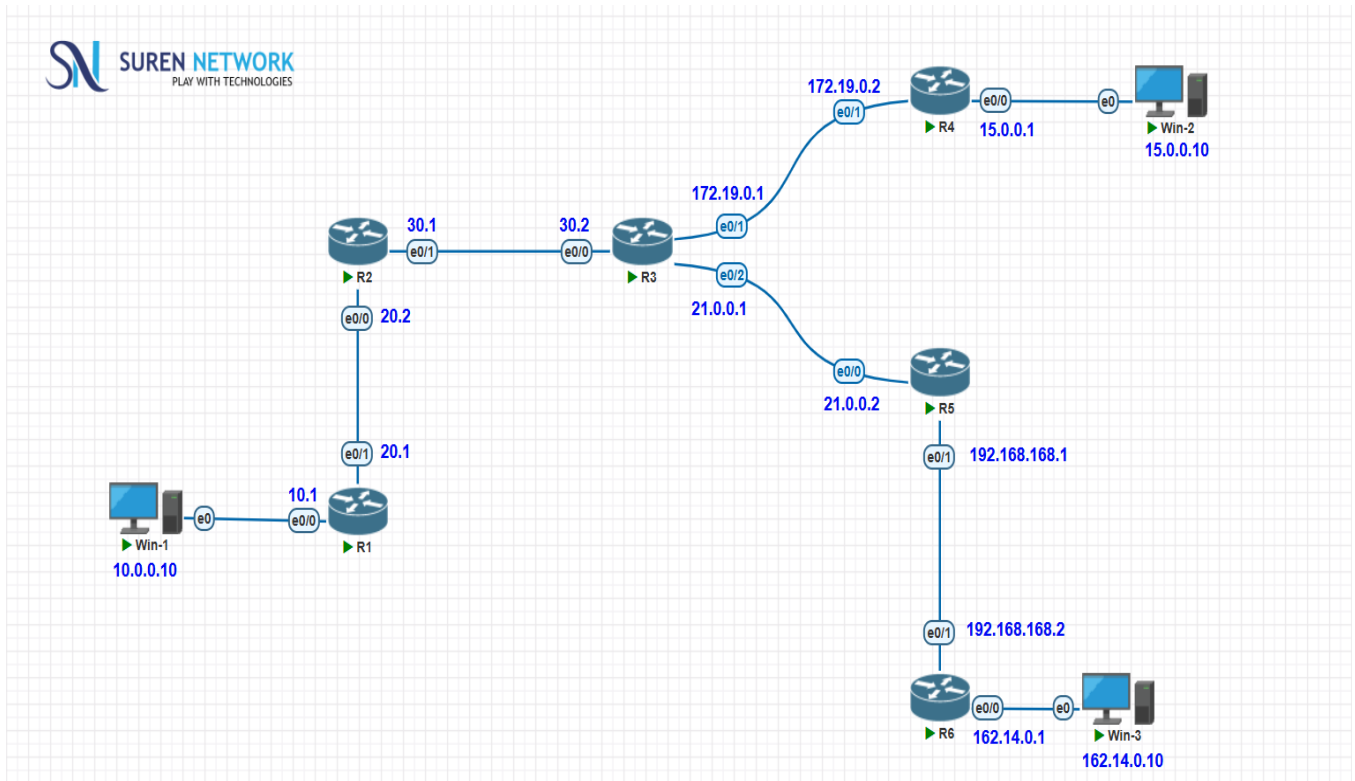


Figure 1: Network topology used for the connectivity testing lab.

This topology contains Win-1, Win-2 and Win-3 connected through routers R1 to R6. The following steps verify end-to-end connectivity between the Windows machines using ICMP ping.

1. Objective

To verify end-to-end network connectivity between the Windows virtual machines using the ping command.

2. IP Addressing

Device	Interface	IP Address
Win-1	e0	10.0.0.10
R1	e0/0	10.0.0.1
R1	e0/1	20.0.0.1
R2	e0/0	20.0.0.2
R2	e0/1	30.0.0.1
R3	e0/0	30.0.0.2
R3	e0/1	172.19.0.1
R3	e0/2	21.0.0.1
R4	e0/1	172.19.0.2
R4	e0/0	15.0.0.1
Win-2	e0	15.0.0.10
R5	e0/0	21.0.0.2
R5	e0/1	192.168.168.1
R6	e0/1	192.168.168.2
R6	e0/0	162.14.0.1
Win-3	e0	162.14.0.10

3. Step 1 – Open Command Prompt on Win-3

Start Win-3 and open Command Prompt using Start → Run → type **cmd** → Enter.

4. Step 2 – Test Win-3 to Win-1

Run:

```
ping 10.0.0.10
```

Successful replies and 0% packet loss confirm that Win-3 can reach Win-1.

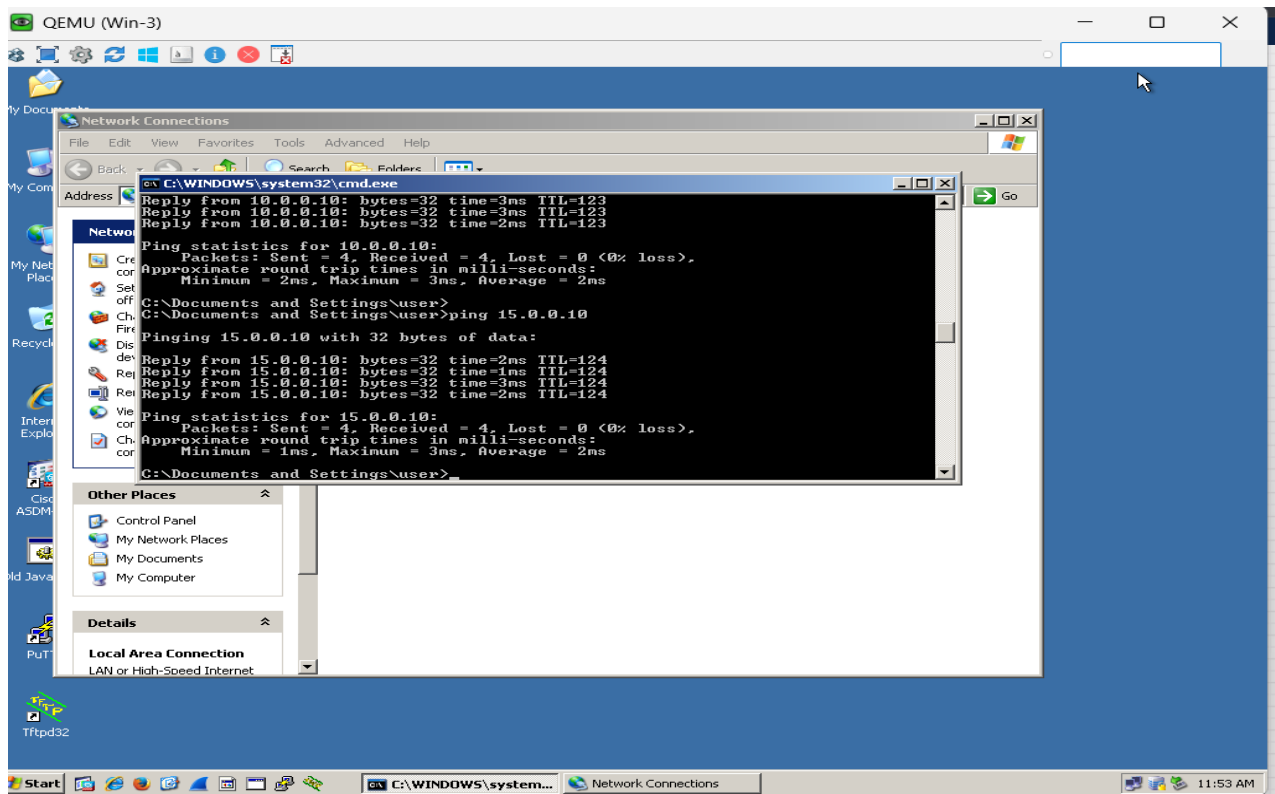


Figure 2: Ping connectivity test.

5. Step 3 – Test Win-3 to Win-2

From Win-3, run:

```
ping 15.0.0.10
```

The captured result shows successful replies from 15.0.0.10 with 0% packet loss.

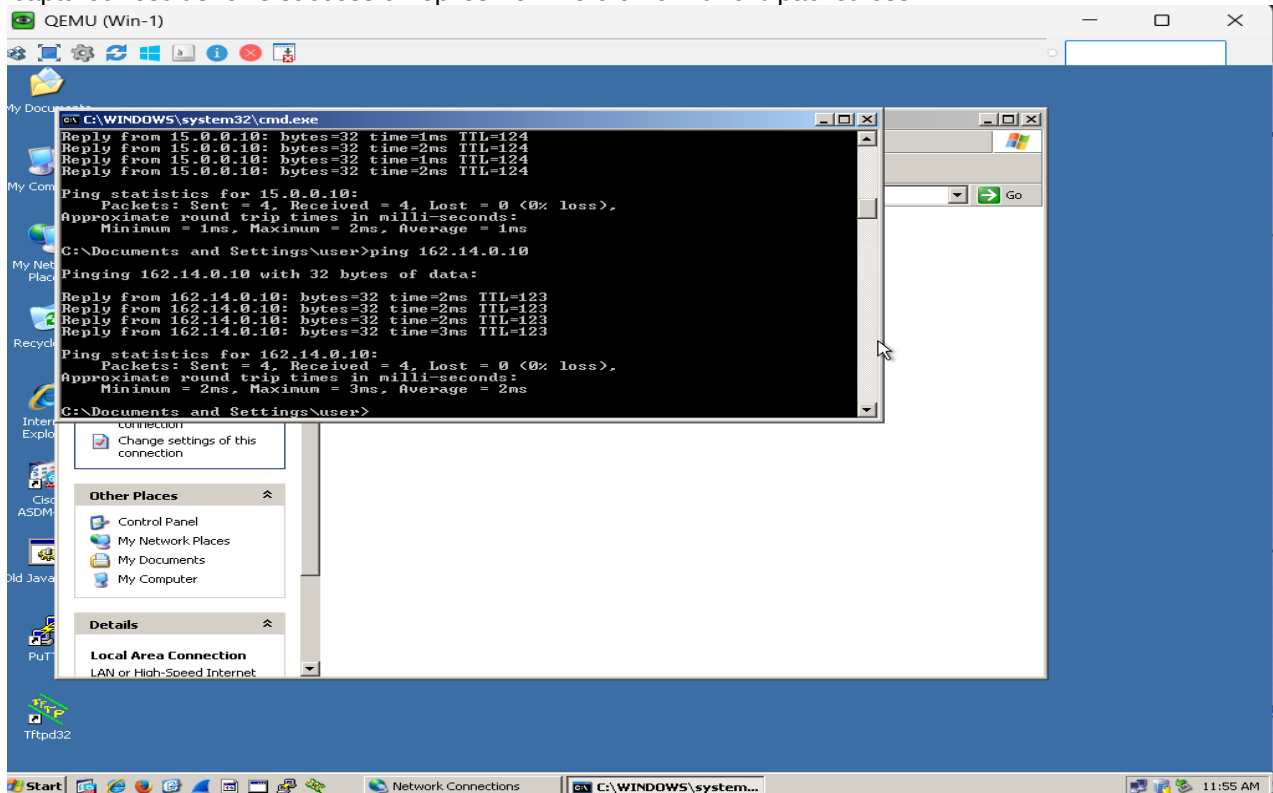


Figure 3: Ping test to Win-2.

6. Step 4 – Test Win-2 to Win-1

On Win-2, open Command Prompt and run:

```
ping 10.0.0.10
```

The test shows successful replies from Win-1 with 0% packet loss.

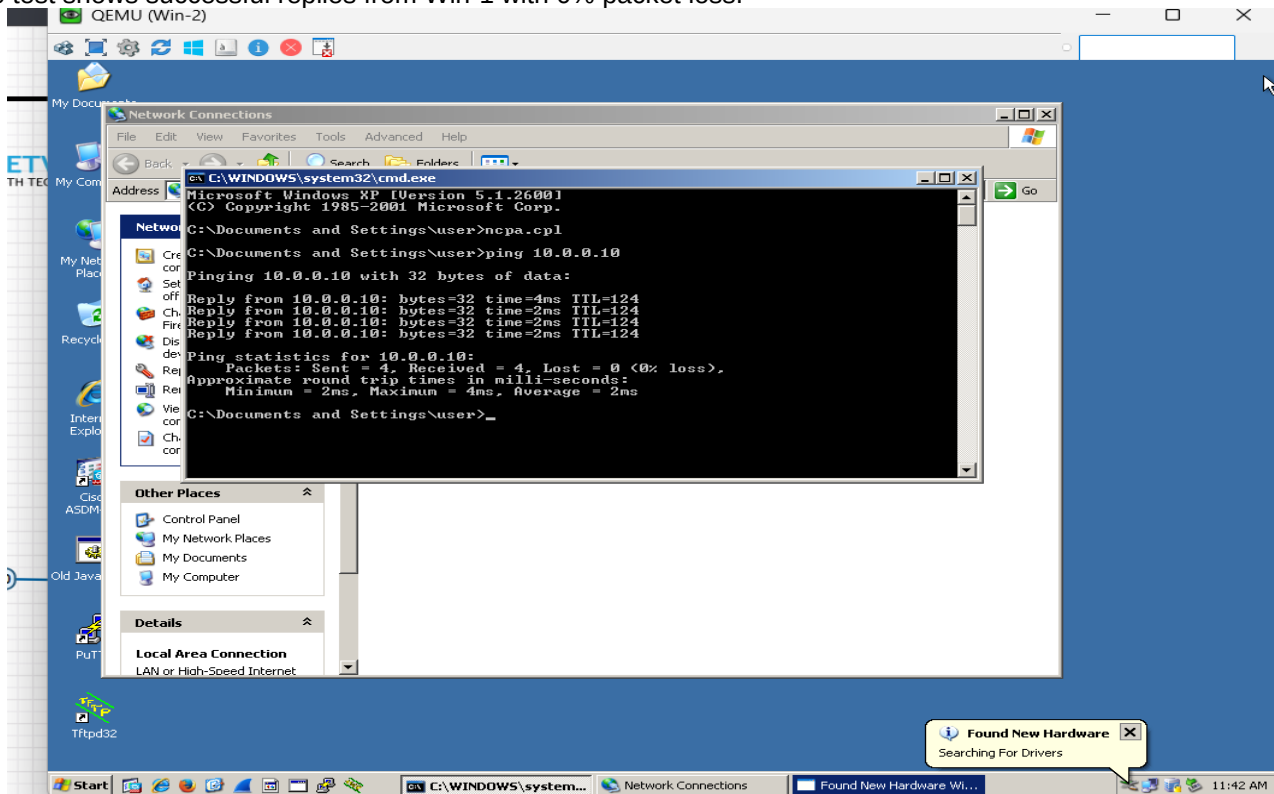


Figure 4: Win-2 to Win-1 ping test.

7. Step 5 – Additional Connectivity Test from Win-1

On Win-1, test Win-3 and Win-2 using their IP addresses:

```
ping 162.14.0.10
```

```
ping 15.0.0.10
```

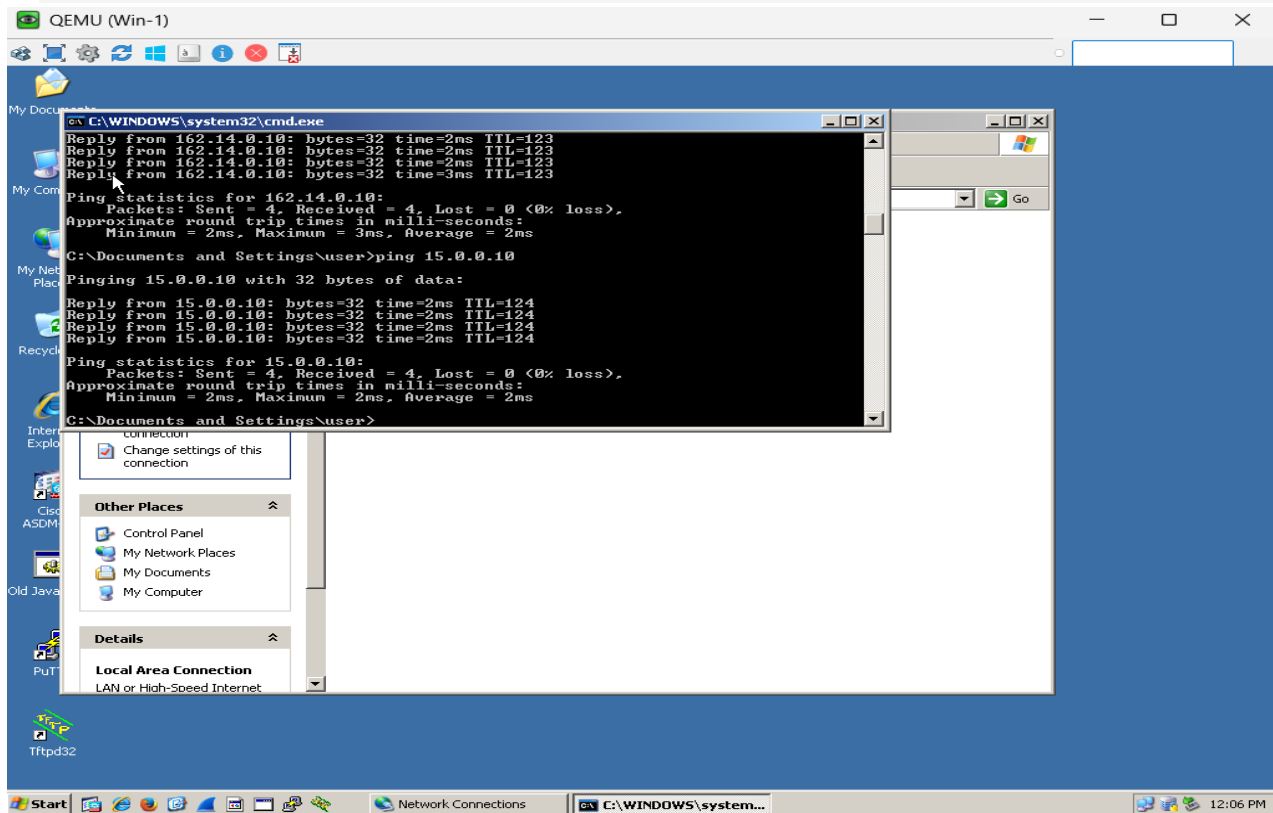


Figure 5: Additional ping verification from Win-1.

8. Verify the Results

Source	Destination	Result
Win-3	10.0.0.10 (Win-1)	Successful / 0% loss
Win-3	15.0.0.10 (Win-2)	Successful / 0% loss
Win-2	10.0.0.10 (Win-1)	Successful / 0% loss
Win-1	162.14.0.10 (Win-3)	Successful / 0% loss
Win-1	15.0.0.10 (Win-2)	Successful / 0% loss

9. Conclusion

The connectivity tests were completed successfully. The Windows machines received ICMP replies with 0% packet loss, confirming that the configured IP addressing and routing paths are functioning correctly.

Note: Ping verifies IP-level connectivity only. It does not confirm that services such as Telnet, SSH, RDP or HTTP are listening.